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Education

- Ph.D. 2012 📖 **Water Resources Engineering**, State University of New York, College of Environmental Science and Forestry (SUNY-ESF) in association with Syracuse University, Syracuse, NY.
- M.S. 2007 📖 **Quaternary Geology**, Lanzhou University, China.
- B.S. 2004 📖 **Geological Science**, Lanzhou University, China.

Appointments

- Dec. 2021 – present 📖 **Team Leader**, Land System Modeling Team, Atmospheric, Climate, & Earth Sciences (ACES) Division, Pacific Northwest National Laboratory, Richland, WA.
- Oct. 2017 – present 📖 **Earth Scientist**, Atmospheric, Climate, & Earth Sciences (ACES) Division, Pacific Northwest National Laboratory, Richland, WA.
- Oct. 2015 – Sep. 2017 📖 **Research Associate**, Atmospheric Sciences & Global Change Division, Pacific Northwest National Laboratory, Richland, WA.
- Oct. 2012 – Oct. 2015 📖 **Research Associate**, Department of Civil and Environmental Engineering, University of Washington, Seattle, WA.
- Aug. 2010 – Aug. 2012 📖 **Research Analyst**, USDA Forest Service Northern Research Station, Syracuse, NY.
- Aug. 2007 – Aug. 2010 📖 **Research Assistant**, Department of Environmental Resources Engineering, SUNY-ESF, Syracuse, NY.

Publications

Refereed Publications

- 61 J.-C. Golaz, W. Lin, X. Zheng, S. Xie, A. F. Roberts, L. P. Van Roekel, P. E. Thornton, A. M. Barthel, A. M. Bradley, J. D. Wolfe, C. Zhang, K. Zhang, S. Zhang, X. S. Asay-Davis, C. B. Begeman, G. Bisht, S. M. Burrows, C.-C.-J. Chen, Y. Feng, E. C. Hunke, et al., “The energy exascale earth system model version 3: 2. overview of the coupled system,” *Journal of Advances in Modeling Earth Systems*, vol. 18, no. 4, Apr. 2026, ISSN: 1942-2466. 🔗 DOI: 10.1029/2025ms005302.
- 60 D. Feng, Z. Tan, D. Engwirda, J. D. Wolfe, D. Xu, C. Liao, G. Bisht, J. J. Benedict, **T. Zhou**, M. Deb, H.-Y. Li, and L. R. Leung, “Disentangling atmospheric, hydrological, and coupling uncertainties in compound flood modeling within a coupled earth system model,” *Natural Hazards and Earth System Sciences*, vol. 25, no. 9, pp. 3619–3639, 2025, ISSN: 1684-9981. 🔗 DOI: 10.5194/nhess-25-3619-2025.
- 59 C. Liao, D. Xu, M. G. Cooper, **T. Zhou**, D. Engwirda, Z. Tan, G. Bisht, H.-Y. Li, L. Li, D. Feng, and L. R. Leung, “Evaluation of flow routing on the unstructured voronoi meshes in earth system modeling,” *Journal of Advances in Modeling Earth Systems*, vol. 17, no. 5, May 2025, ISSN: 1942-2466. 🔗 DOI: 10.1029/2024ms004737.
- 58 K. C. Solander, **T. Zhou**, K. E. Bennett, and J. Schwenk, “Evaluation of cmip6 streamflow in the arctic,” *Journal of Hydrometeorology*, vol. 26, no. 10, pp. 1475–1492, Oct. 2025, ISSN: 1525-7541. 🔗 DOI: 10.1175/jhm-d-24-0124.1.

- 57 H. Tran, **T. Zhou**, Z. Tan, Y. Fang, and L. Ruby Leung, "Improving the prediction of daily reservoir releases over the conus using conditioned lstm," *Journal of Hydrology*, vol. 661, p. 133 750, Nov. 2025, ISSN: 0022-1694. [DOI: 10.1016/j.jhydrol.2025.133750](#).
- 56 Y. Yao, A. Ducharne, B. I. Cook, S. J. De Hertog, K. S. Aas, P. F. Arboleda-Obando, J. Buzan, J. Colin, M. Costantini, B. Decharme, D. M. Lawrence, P. Lawrence, L. R. Leung, M.-H. Lo, N. Devaraju, W. R. Wieder, R.-J. Wu, **T. Zhou**, J. Jägermeyr, S. McDermid, et al., "Impacts of irrigation expansion on moist-heat stress based on irrmip results," *Nature Communications*, vol. 16, no. 1, Jan. 2025, ISSN: 2041-1723. [DOI: 10.1038/s41467-025-56356-1](#).
- 55 Y. Yao, W. Thiery, A. Ducharne, B. I. Cook, A. Ding, S. J. De Hertog, P. Sieber, K. S. Aas, P. F. Arboleda-Obando, J. Colin, M. Costantini, B. Decharme, D. M. Lawrence, P. Lawrence, L. R. Leung, M.-H. Lo, N. Devaraju, R.-J. Wu, **T. Zhou**, J. Jägermeyr, et al., "Irrigation-induced land water depletion aggravated by climate change," *Nature Water*, vol. 3, no. 12, pp. 1424–1435, Nov. 2025, ISSN: 2731-6084. [DOI: 10.1038/s44221-025-00529-1](#).
- 54 D. Feng, Z. Tan, D. Engwirda, J. D. Wolfe, D. Xu, C. Liao, G. Bisht, J. J. Benedict, **T. Zhou**, H.-Y. Li, and L. R. Leung, "Simulation of compound flooding using river-ocean two-way coupled e3sm ensemble on variable-resolution meshes," *Journal of Advances in Modeling Earth Systems*, vol. 16, no. 6, 2024, ISSN: 1942-2466. [DOI: 10.1029/2023ms004054](#).
- 53 T. B. Thurber, D. Broman, **T. Zhou**, and N. Voisin, "Wmpy-power: A python package for process-based regional hydropower simulation," *Journal of Open Source Software*, vol. 9, no. 103, p. 7225, Nov. 2024, ISSN: 2475-9066. [DOI: 10.21105/joss.07225](#).
- 52 H. Tran, Y. Fang, Z. Tan, **T. Zhou**, and L. R. Leung, "Quantifying the impacts of land-cover change on the hydrologic response to hurricane ida in the lower mississippi river basin," *Journal of Hydrometeorology*, vol. 25, no. 6, pp. 899–914, 2024, ISSN: 1525-7541. [DOI: 10.1175/jhm-d-23-0094.1](#).
- 51 D. Xu, G. Bisht, Z. Tan, C. Liao, **T. Zhou**, H.-Y. Li, and L. R. Leung, "Disentangling the hydrological and hydraulic controls on streamflow variability in energy exascale earth system model (e3sm) v2 – a case study in the pantanal region," *Geoscientific Model Development*, vol. 17, no. 3, pp. 1197–1215, Feb. 2024, ISSN: 1991-9603. [DOI: 10.5194/gmd-17-1197-2024](#).
- 50 D. Xu, G. Bisht, Z. Tan, E. Sinha, A. V. Di Vittorio, **T. Zhou**, V. Y. Ivanov, and L. R. Leung, "Climate change will reduce north american inland wetland areas and disrupt their seasonal regimes," *Nature Communications*, vol. 15, no. 1, Mar. 2024, ISSN: 2041-1723. [DOI: 10.1038/s41467-024-45286-z](#).
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- 48 M. G. Cooper and **T. Zhou**, "Baseflow: A matlab and gnu octave package for baseflow recession analysis," *Journal of Open Source Software*, vol. 8, no. 90, p. 5492, Oct. 2023, ISSN: 2475-9066. [DOI: 10.21105/joss.05492](#).
- 47 B. E. Harrop, K. Balaguru, J.-C. Golaz, L. R. Leung, S. Mahajan, A. M. Rhoades, P. A. Ullrich, C. Zhang, X. Zheng, **T. Zhou**, P. M. Caldwell, N. D. Keen, and A. Mametjanov, "Evaluating the water cycle over conus at the watershed scale for the energy exascale earth system model version 1 (e3smv1) across resolutions," *Journal of Advances in Modeling Earth Systems*, vol. 15, no. 11, Nov. 2023, ISSN: 1942-2466. [DOI: 10.1029/2022ms003490](#).
- 46 C. Liao, **T. Zhou**, D. Xu, M. G. Cooper, D. Engwirda, H.-Y. Li, and L. R. Leung, "Topological relationship-based flow direction modeling: Mesh-independent river networks representation," *Journal of Advances in Modeling Earth Systems*, vol. 15, no. 2, Feb. 2023, ISSN: 1942-2466. [DOI: 10.1029/2022ms003089](#).
- 45 C. Liao, **T. Zhou**, D. Xu, Z. Tan, G. Bisht, M. G. Cooper, D. Engwirda, H.-Y. Li, and L. R. Leung, "Topological relationship-based flow direction modeling: Stream burning and depression filling,"

- 44 B. Pachev, L. R. Leung, **T. Zhou**, and C. Dawson, “One-way coupling of e3sm with adcirc demonstrated on hurricane harvey,” *Natural Hazards*, vol. 119, no. 3, pp. 2063–2087, 2023, ISSN: 1573-0840.  DOI: 10.1007/s11069-023-06192-7.
- 43 S. J. Silva, S. M. Burrows, K. Calvin, P. J. Cameron-Smith, X. Shi, and **T. Zhou**, “Contrasting the biophysical and radiative effects of rising CO_2 concentrations on ozone dry deposition fluxes,” *Journal of Geophysical Research: Atmospheres*, vol. 128, no. 6, Mar. 2023, ISSN: 2169-8996.  DOI: 10.1029/2022jd037668.
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- 41 **T. Zhou**, S.-C. Kao, W. Xu, S. Gangrade, and N. Voisin, “Impacts of climate change on subannual hydropower generation: A multi-model assessment of the united states federal hydropower plant,” *Environmental Research Letters*, vol. 18, no. 3, p. 034 009, Feb. 2023, ISSN: 1748-9326.  DOI: 10.1088/1748-9326/acb58d.
- 40 H. Eldardiry, **T. Zhou**, M. Huang, and O. S. Chegwiddden, “The role of groundwater withdrawals on river regulation: Example from the columbia river basin,” *Water Resources Research*, vol. 58, no. 6, 2022, ISSN: 1944-7973.  DOI: 10.1029/2020wr028955.
- 39 D. Feng, Z. Tan, D. Engwirda, C. Liao, D. Xu, G. Bisht, **T. Zhou**, H.-Y. Li, and L. R. Leung, “Investigating coastal backwater effects and flooding in the coastal zone using a global river transport model on an unstructured mesh,” *Hydrology and Earth System Sciences*, vol. 26, no. 21, pp. 5473–5491, Nov. 2022, ISSN: 1607-7938.  DOI: 10.5194/hess-26-5473-2022.
- 38 J.-C. Golaz, L. P. Van Roekel, X. Zheng, A. F. Roberts, J. D. Wolfe, W. Lin, A. M. Bradley, Q. Tang, M. E. Maltrud, R. M. Forsyth, C. Zhang, **T. Zhou**, K. Zhang, C. S. Zender, M. Wu, H. Wang, A. K. Turner, B. Singh, J. H. Richter, Y. Qin, et al., “The doe e3sm model version 2: Overview of the physical model and initial model evaluation,” *Journal of Advances in Modeling Earth Systems*, vol. 14, no. 12, Dec. 2022, ISSN: 1942-2466.  DOI: 10.1029/2022ms003156.
- 37 S.-C. Kao, M. Ashfaq, D. Rastogi, S. Gangrade, R. Uria Martinez, A. Fernandez, G. Konapala, N. Voisin, **T. Zhou**, W. Xu, H. Gao, B. Zhao, and G. Zhao, *The Third Assessment of the Effects of Climate Change on Federal Hydropower*. 2022.  DOI: 10.2172/1887712.
- 36 H.-Y. Li, Z. Tan, H. Ma, Z. Zhu, G. W. Abeshu, S. Zhu, S. Cohen, **T. Zhou**, D. Xu, and L. R. Leung, “A new large-scale suspended sediment model and its application over the united states,” *Hydrology and Earth System Sciences*, vol. 26, no. 3, pp. 665–688, Feb. 2022, ISSN: 1607-7938.  DOI: 10.5194/hess-26-665-2022.
- 35 C. Liao, **T. Zhou**, D. Xu, R. Barnes, G. Bisht, H.-Y. Li, Z. Tan, T. Tesfa, Z. Duan, D. Engwirda, and L. R. Leung, “Advances in hexagon mesh-based flow direction modeling,” *Advances in Water Resources*, vol. 160, p. 104 099, Feb. 2022, ISSN: 0309-1708.  DOI: 10.1016/j.adwatres.2021.104099.
- 34 N. Sun, H. Yan, M. S. Wigmosta, J. Lundquist, S. Dickerson-Lange, and **T. Zhou**, “Forest canopy density effects on snowpack across the climate gradients of the western united states mountain ranges,” *Water Resources Research*, vol. 58, no. 1, Jan. 2022, ISSN: 1944-7973.  DOI: 10.1029/2020wr029194.
- 33 D. Xu, G. Bisht, **T. Zhou**, L. R. Leung, and M. Pan, “Development of land-river two-way hydrologic coupling for floodplain inundation in the energy exascale earth system model,” *Journal of Advances in Modeling Earth Systems*, vol. 14, no. 8, Aug. 2022, ISSN: 1942-2466.  DOI: 10.1029/2021ms002772.

- 32 C. Zhang, J.-C. Golaz, R. Forsyth, T. Vo, S. Xie, Z. Shaheen, G. L. Potter, X. S. Asay-Davis, C. S. Zender, W. Lin, C.-C. Chen, C. R. Terai, S. Mahajan, **T. Zhou**, K. Balaguru, Q. Tang, C. Tao, Y. Zhang, T. Emmenegger, S. Burrows, et al., "The e3sm diagnostics package (e3sm diags v2.7): A python-based diagnostics package for earth system model evaluation," *Geoscientific Model Development*, vol. 15, no. 24, pp. 9031–9056, Dec. 2022, ISSN: 1991-9603. [DOI: 10.5194/gmd-15-9031-2022](#).
- 31 X. Zheng, Q. Li, **T. Zhou**, Q. Tang, L. P. Van Roekel, J.-C. Golaz, H. Wang, and P. Cameron-Smith, "Description of historical and future projection simulations by the global coupled e3smv1.0 model as used in cmip6," *Geoscientific Model Development*, vol. 15, no. 9, pp. 3941–3967, May 2022, ISSN: 1991-9603. [DOI: 10.5194/gmd-15-3941-2022](#).
- 30 Y. Cheng, M. Huang, B. Zhu, G. Bisht, **T. Zhou**, Y. Liu, F. Song, and X. He, "Validation of the community land model version 5 over the contiguous united states (conus) using in situ and remote sensing data sets," *Journal of Geophysical Research: Atmospheres*, vol. 126, no. 5, Mar. 2021, ISSN: 2169-8996. [DOI: 10.1029/2020jd033539](#).
- 29 S. M. Burrows, M. Maltrud, X. Yang, Q. Zhu, N. Jeffery, X. Shi, D. Ricciuto, S. Wang, G. Bisht, J. Tang, J. Wolfe, B. E. Harrop, B. Singh, L. Brent, S. Baldwin, **T. Zhou**, P. Cameron-Smith, N. Keen, N. Collier, M. Xu, et al., "The doe e3sm v1.1 biogeochemistry configuration: Description and simulated ecosystem-climate responses to historical changes in forcing," *Journal of Advances in Modeling Earth Systems*, vol. 12, no. 9, 2020, ISSN: 1942-2466. [DOI: 10.1029/2019ms001766](#).
- 28 N. Voisin, A. Dyreson, T. Fu, M. O'Connell, S. W. Turner, **T. Zhou**, and J. Macknick, "Impact of climate change on water availability and its propagation through the western u.s. power grid," *Applied Energy*, vol. 276, p. 115 467, Oct. 2020, ISSN: 0306-2619. [DOI: 10.1016/j.apenergy.2020.115467](#).
- 27 **T. Zhou** and T. Endreny, "The straightening of a river meander leads to extensive losses in flow complexity and ecosystem services," *Water*, vol. 12, no. 6, p. 1680, 2020, ISSN: 2073-4441. [DOI: 10.3390/w12061680](#).
- 26 **T. Zhou**, L. R. Leung, G. Leng, N. Voisin, H.-Y. Li, A. P. Craig, T. Tesfa, and Y. Mao, "Global irrigation characteristics and effects simulated by fully coupled land surface, river, and water management models in e3sm," *Journal of Advances in Modeling Earth Systems*, vol. 12, no. 10, Oct. 2020, ISSN: 1942-2466. [DOI: 10.1029/2020ms002069](#).
- 25 P. M. Caldwell, A. Mametjanov, Q. Tang, L. P. Van Roekel, J.-C. Golaz, W. Lin, D. C. Bader, N. D. Keen, Y. Feng, R. Jacob, M. E. Maltrud, A. F. Roberts, M. A. Taylor, M. Veneziani, H. Wang, J. D. Wolfe, K. Balaguru, P. Cameron-Smith, L. Dong, S. A. Klein, et al., "The doe e3sm coupled model version 1: Description and results at high resolution," *Journal of Advances in Modeling Earth Systems*, vol. 11, no. 12, pp. 4095–4146, Dec. 2019, ISSN: 1942-2466. [DOI: 10.1029/2019ms001870](#).
- 24 J.-C. Golaz, P. M. Caldwell, L. P. Van Roekel, M. R. Petersen, Q. Tang, J. D. Wolfe, G. Abeshu, V. Anantharaj, X. S. Asay-Davis, D. C. Bader, S. A. Baldwin, G. Bisht, P. A. Bogenschutz, M. Branstetter, M. A. Brunke, S. R. Brus, S. M. Burrows, P. J. Cameron-Smith, A. S. Donahue, M. Deakin, et al., "The doe e3sm coupled model version 1: Overview and evaluation at standard resolution," *Journal of Advances in Modeling Earth Systems*, vol. 11, no. 7, pp. 2089–2129, 2019, ISSN: 1942-2466. [DOI: 10.1029/2018ms001603](#).
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- 22 H. L. Shah, **T. Zhou**, M. Huang, and V. Mishra, "Strong influence of irrigation on water budget and land surface temperature in indian subcontinental river basins," *Journal of Geophysical Research: Atmospheres*, vol. 124, no. 3, pp. 1449–1462, Feb. 2019, ISSN: 2169-8996. [DOI: 10.1029/2018jd029132](#).
- 21 H. L. Shah, **T. Zhou**, N. Sun, M. Huang, and V. Mishra, "Roles of irrigation and reservoir operations in modulating terrestrial water and energy budgets in the indian subcontinental river basins," *Journal of*

- 20 J. Bao, **T. Zhou**, M. Huang, Z. Hou, W. Perkins, S. Harding, S. Titzler, G. Hammond, H. Ren, P. Thorne, S. Suffield, C. Murray, and J. Zachara, “Modulating factors of hydrologic exchanges in a large-scale river reach: Insights from three-dimensional computational fluid dynamics simulations,” *Hydrological Processes*, vol. 32, no. 23, pp. 3446–3463, Oct. 2018, ISSN: 1099-1085. [DOI: 10.1002/hyp.13266](#).
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- 13 **T. Zhou**, N. Voisin, G. Leng, M. Huang, and I. Kraucunas, “Sensitivity of regulated flow regimes to climate change in the western united states,” *Journal of Hydrometeorology*, vol. 19, no. 3, pp. 499–515, Mar. 2018, ISSN: 1525-7541. [DOI: 10.1175/jhm-d-17-0095.1](#).
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- 3 **T. Zhou**, B. Pan, X. Liu, H. Su, and Z. Hu, "The discovery of ice-wedge casts in Ordos Plateau, China and permafrost boundary establishment," *Journal of Glaciology and Geocryology*, 2008, in Chinese with English abstract. [URL: http://bcdt.westgis.ac.cn/CN/abstract/abstract415.shtml](http://bcdt.westgis.ac.cn/CN/abstract/abstract415.shtml).
- 2 P. Baotian, S. Huai, H. Chunsheng, H. Ziaofei, Z. Tian, and L. Jijun, "Discovery of a 1.0 ma yellow river terrace and redating of the fourth yellow river terrace in lanzhou area," *Progress in Natural Science*, vol. 17, no. 2, pp. 197–205, Feb. 2007, ISSN: 1002-0071. [DOI: 10.1080/10020070612331343246](#).
- 1 B. Pan, H. Su, X. Liu, X. Hu, **T. Zhou**, C. Hu, and J. Li, "River terraces of the Yellow River and their genesis in eastern Lanzhou Basin during last 1.2 Ma," *Quaternary Sciences*, 2007, in Chinese with English abstract. [URL: http://www.dsyyj.com.cn/CN/abstract/abstract9193.shtml](http://www.dsyyj.com.cn/CN/abstract/abstract9193.shtml).

Publications In Review / In Press

- C. Liao, D. Engwirda, D. Xu, **T. Zhou**, Z. Tan, and L. R. Leung, "An unstructured voronoi mesh framework for large scale hydrologic and land surface models," *Journal of Advances in Modeling Earth Systems*, 2026, accepted.
- Z. Tan, D. Xu, **T. Zhou**, D. Feng, L. Li, M. Cooper, C. Liao, G. Bisht, G. Abeshu, H.-Y. Li, A. Kassam, and L. R. Leung, "The past and future changes of river sediment in the u.s. mid-atlantic," *Earth's Future*, 2026, in review.
- D. Xu, Z. Tan, E. Fluet-Chouinard, G. Bisht, D. Hao, D. Feng, **T. Zhou**, et al., "Muted global changes despite large regional responses: The contrasting future trajectories of inundated and saturated wetlands," *Nature Climate Change*, 2026, in review.
- **T. Zhou**, Y. Qian, and L. R. Leung, "Can we trust llms for complex earth system model analysis? silent failure and evidence from module-grounded benchmarking," Apr. 2026. [DOI: 10.5194/egusphere-2026-2237](#).

Data Publications

- **T. Zhou**, S.-C. Kao, W. Xu, S. Gangrade, and N. Voisin, *Multi-model hydropower projections for the united states federal power marketing areas under cmip5 climate change conditions*, 2022. [DOI: 10.5281/ZENODO.6506088](#).
- N. Voisin, A. Dyreson, T. Fu, M. O'Connell, S. Turner, **T. Zhou**, and J. Macknick, *Plexos inputs for voisin et al. 2020 [data set]*, en, 2020. [DOI: 10.25584/DATA.2020-06.1318/1635205](#).

- S. Gosling, H. Müller Schmied, R. Betts, J. Chang, P. Ciais, R. Dankers, P. Döll, S. Eisner, M. Flörke, D. Gerten, M. Grillakis, N. Hanasaki, S. Hagemann, M. Huang, Z. Huang, S. Jerez, H. Kim, A. Koutroulis, G. Leng, X. Liu, et al., *Isimip2a simulation data from water (global) sector*, en, 2017. [DOI: 10.5880/PIK.2017.010](https://doi.org/10.5880/PIK.2017.010).

Non-Refereed Publications

- D. J. Nowak, R. E. I. Hoehn, A. R. Bodine, E. J. Greenfield, A. Ellis, T. A. Endreny, Y. Yang, **T. Zhou**, and R. Henry, *Assessing urban forest effects and values: Toronto's urban forest*. 2013. [DOI: 10.2737/nrs-rb-79](https://doi.org/10.2737/nrs-rb-79).

Selected Presentations

- 2025
 - **Zhou T.** and C. Liao. *River Delta Processes in Earth System Models – A New Bifurcation Scheme*. American Geophysical Union Fall Meeting, New Orleans, LA.
- 2024
 - **Zhou T.** (invited lecture) *River Modeling in E3SM*. 2024 E3SM Tutorial Workshop, National Energy Research Scientific Computing Center, Berkeley, CA.
- 2023
 - **Zhou T.**, M. Cooper, C. Liao, D. Xu, D. Engwirda, N. Sun, Z. Tan, H-Y. Li, D. Feng, G. Bisht, L. Leung. *Modeling Inter-Basin Water Transfer in E3SM: A Delaware River Basin Case Study*. American Geophysical Union Fall Meeting, San Francisco, CA.
 - Liao C., **T. Zhou**, D. Engwirda, D. Xu, M.G. Cooper, Z. Tan, G. Bisht, H-Y. Li, L. Leung. *Evaluation of river routing on an unstructured mesh in E3SM*. American Geophysical Union Fall Meeting, San Francisco, CA.
 - **Zhou T.** (invited lecture) *Hydroelectricity Modeling*. Global Climate: Physical Modeling (GCEE6320), University of Houston, Houston, TX.
 - **Zhou T.** (invited talk) *Irrigation modeling in Energy Exascale Earth System Model (E3SM)*. Aspen Global Change Institute Workshop: Irrigation in the Earth System, Aspen, CO.
- 2020
 - **Zhou T.**, L. Leung, and D. Xu. *Changes in flood characteristics in the future*. American Geophysical Union Fall Meeting.
 - Eldardiry H.A., **T. Zhou**, and M. Huang. *The Role of Groundwater Withdrawals on River Regulation: Example from the Columbia River Basin*. American Meteorological Society Meeting, Boston, MA.
- 2018
 - **Zhou T.**, R. Leung, N. Voisin, H. Li, G. Leng, T. Tesfa. *Global irrigation water withdrawal simulated by fully coupled land surface, river, and water management models*. American Geophysical Union Fall Meeting, Washington, DC.
- 2017
 - **Zhou T.**, N. Voisin, T. Fu. *Non-stationary hydropower generation projection over the western United States*. American Geophysical Union Fall Meeting, New Orleans, LA.
- 2016
 - **Zhou T.**, J. Bao, M. Huang, Z. Hou, E. Arntzen, R. Mackley. *Quantifying hyporheic exchange dynamics in a highly regulated large river reach*. American Geophysical Union Fall Meeting, San Francisco, CA.
- 2014
 - **Zhou T.**, B. Nijssen, I. Haddeland, H. Gao, D.P. Lettenmaier. *Reservoir in Global Water Cycle: Macro Scale Hydrologic Modeling for Water Management*. American Geophysical Union Fall Meeting, San Francisco, CA.

Selected Presentations (continued)

- 2013
- **Zhou T.**, B. Nijssen, I. Haddeland, D.P. Lettenmaier. *Macro Scale Hydrologic Modeling for Water Management: Re-construction of Large Reservoir Storage Time Series in the Continental U.S.* American Geophysical Union Fall Meeting, San Francisco, CA.
 - Lettenmaier D.P., **T. Zhou**, G.J. Huffman, B. Nijssen. *Evaluation of TMPA v7 Real-Time Precipitation for Global Hydrologic Prediction.* 3rd International Workshop on Global Flood Monitoring & Modelling, College Park, MD.
- 2012
- **Zhou T.**, A.S. Ward, B.L. O'Connor, T.A. Endreny. *Floodplain Hyporheic Response under Dam Release Hydrographs.* American Geophysical Union Fall Meeting, San Francisco, CA.
- 2011
- **Zhou T.**, T.A. Endreny. *Hydrodynamic impacts of disrupting point bar steering with river restoration structures.* American Geophysical Union Fall Meeting, San Francisco, CA.
 - **Zhou T.**, T.A. Endreny. *Changes of hydraulic patterns with in-channel restoration structures at a point bar.* 11th Annual Meeting of the American Ecological Engineering Society, Asheville, NC.
- 2010
- **Zhou T.**, T.A. Endreny. *Flume analysis of in-channel restoration structures and impacts to secondary circulation flows.* World Congress of the International Commission of Agricultural and Biosystems Engineering, Quebec City, Canada.
 - **Zhou T.**, T.A. Endreny. *Hyporheic exchange flow around in-channel restoration structures: simulation and flume experiments.* Association of American Geographers Annual Meeting, Washington, DC.
- 2009
- **Zhou T.**, T.A. Endreny. (invited talk) *Hydraulic impacts of in-channel restoration structures in a meander band: simulation with CFD.* USGS / FL-LOWPA / Finger Lakes Institute Workshop on Techniques for Evaluating Water Resources in the Finger Lakes, Geneva, NY.

Service

Editorial	■ Editorial board member, <i>Advances in Climate Change Research</i> (2018 – present). Editorial board member, <i>Journal of the American Water Resources Association (JAWRA)</i> (2024 – present).
Review Panels	■ Review panelist, 2014 US EPA National Priorities Grant; 2022 NOAA NWS Office of Science and Technology Integration: Unified Forecast System Grant.
Sessions	■ AGU 2020 session organizer.
Journal Refereeing	■ <i>Water Resources Research</i> ; <i>Journal of Hydrometeorology</i> ; <i>Journal of Hydrology</i> ; <i>Journal of Geophysical Research: Atmospheres</i> ; <i>Hydrology and Earth System Sciences</i> ; <i>Hydrological Processes</i> ; <i>Hydrogeology Journal</i> ; <i>AGU Books</i> ; <i>Earth System Dynamics</i> ; <i>Water Science and Technology</i> ; <i>International Journal of Climatology</i> ; <i>Remote Sensing</i> ; <i>Estuarine, Coastal and Shelf Science</i> ; <i>WIREs Water</i> ; among others.

Honors, Awards, & License

Honors and Awards

- 2018 ■ PNNL Exceptional Contribution Program (ECP) Award.
- 2017 ■ EBSD Best Award, Pacific Northwest National Laboratory.
- 2011 ■ AEES Conference Travel Grant.
- 2007 – 2010 ■ Tuition Scholarship Award, SUNY-ESF, NY.
- 2004 ■ First Place, 5th “Challenge Cup” of the College Student Research Competition of Gansu Province, China.

Honors, Awards, & License (continued)

2002  Tuition Scholarship Award, Lanzhou University, China.

Professional License

E.I.T.  Washington, License No. 34995.

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